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WO-500-02 — Ultrasonic thickness survey of 6"-P300-LS-2003

Area	A500 Pipe rack	**Task**	UT Ultrasonic thickness survey
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	rack-bent:1200 — Pipe rack bent at E 100'-0" — E 100'-0" N 0.000" EL 122'-0"		
Equipment	—	**Lines**	6"-P300-LS-2003
Isometrics	ISO-6"-P300-LS-2003 (generated); owner's drawings ISO-LS-2003		
Permits	General work permit		

Reason

Thickness survey of the steam header.

Scope of work

1. Erect access and remove insulation at the TMLs where fitted (class S300, insulated).
2. Calibrate the thickness gauge on a step block covering 0.21875 in.
3. Take four readings 90° apart at each of the 23 TMLs (SW-WO-6).
4. Report readings against the owner's minimum wall; raise a spool replacement for any reading below it.

Affected items: 6"-P300-LS-2003/T:2"-P300-LS-3007, 6"-P300-LS-2003/p4, 6"-P300-LS-2003/T:1"-P300-HS-5003, 6"-P300-LS-2003/p6, 6"-P300-LS-2003/T:2"-P300-LS-3009, 6"-P300-LS-2003/p8, 6"-P300-LS-2003/T:1"-P300-HS-5008, 6"-P300-LS-2003/p10, 6"-P300-LS-2003/T:1"-P300-HS-5013, 6"-P300-LS-2003/p12, 6"-P300-LS-2003/T:1"-P300-HS-5018, 6"-P300-LS-2003/p14, 6"-P300-LS-2003/T:1"-P300-HS-5023, 6"-P300-LS-2003/p16, 6"-P300-LS-2003/p17, 6"-P300-LS-2003/T:1"-P300-HS-5028, 6"-P300-LS-2003/p19, 6"-P300-LS-2003/T:1"-P300-HS-5033, 6"-P300-LS-2003/p21, 6"-P300-LS-2003/T:1"-P300-HS-5038, 6"-P300-LS-2003/T:3/4"-P150-V-8009, 6"-P300-LS-2003/T:3/4"-P150-D-8008, 6"-P300-LS-2003/T:3/4"-P150-D-8007

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-WO-1, SW-WO-2, SW-WO-6, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-LS-2003	6"-P300-LS-2003	6"-P300-LS-2003	7	0 error(s), 2 warning(s) — see below

ISO-LS-2003 (PDF): ../isometrics-imported/ISO-LS-2003.pdf — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-LS-2003	6	warning	branch weld 3/4"-P150-V-8009/W1 (E 175'-6" N -1'-0" EL 123'-5", sheet 6) has no label
ISO-LS-2003	7	warning	branch weld 3/4"-P150-D-8007/W1 (E 178'-0" N -1'-0" EL 123'-5", sheet 7) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
rack-bent:1200	structure	Pipe rack bent at E 100'-0"		E 100'-0" N 0.000" EL 122'-0"
6"-P300-LS-2003/T:2"-P300-LS-3007	item	TEE REDUCING 6" × 2" P300	6"-P300-LS-2003	E 22'-0" N -12" EL 123'-0"
6"-P300-LS-2003/p4	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 23'-9" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5003	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 25'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p6	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 30'-6" N -12" EL 123'-0"
6"-P300-LS-2003/T:2"-P300-LS-3009	item	TEE REDUCING 6" × 2" P300	6"-P300-LS-2003	E 35'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p8	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 40'-6" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5008	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 45'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p10	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 55'-6" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5013	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 65'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p12	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 75'-0" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5018	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 84'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p14	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 95'-0" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5023	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 105'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p16	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 115'-8-1/2" N -12" EL 123'-0"
6"-P300-LS-2003/p17	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 135'-3-1/2" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5028	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 145'-6" N -12" EL 123'-0"
6"-P300-LS-2003/p19	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 155'-6" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5033	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 165'-6" N -12" EL 123'-0"

Tag	Kind	Description	Line	Location
6"-P300-LS-2003/p21	item	PIPE 6" P300 BEVELLED ENDS	6"-P300-LS-2003	E 170'-0" N -12" EL 123'-0"
6"-P300-LS-2003/T:1"-P300-HS-5038	item	TEE REDUCING 6" × 1" P300	6"-P300-LS-2003	E 174'-6" N -12" EL 123'-0"
6"-P300-LS-2003/T:3/4"-P150-V-8009	item	TEE REDUCING 6" × 3/4" P300	6"-P300-LS-2003	E 175'-6" N -12" EL 123'-0"
6"-P300-LS-2003/T:3/4"-P150-D-8008	item	TEE REDUCING 6" × 3/4" P300	6"-P300-LS-2003	E 177'-0" N -12" EL 123'-0"
6"-P300-LS-2003/T:3/4"-P150-D-8007	item	TEE REDUCING 6" × 3/4" P300	6"-P300-LS-2003	E 178'-0" N -12" EL 123'-0"

Inspections

Id	Inspection	Criteria	Basis
WO-500-02-I1	Ultrasonic thickness at 23 TMLs, four readings each	Nominal wall 0.21875 in; each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

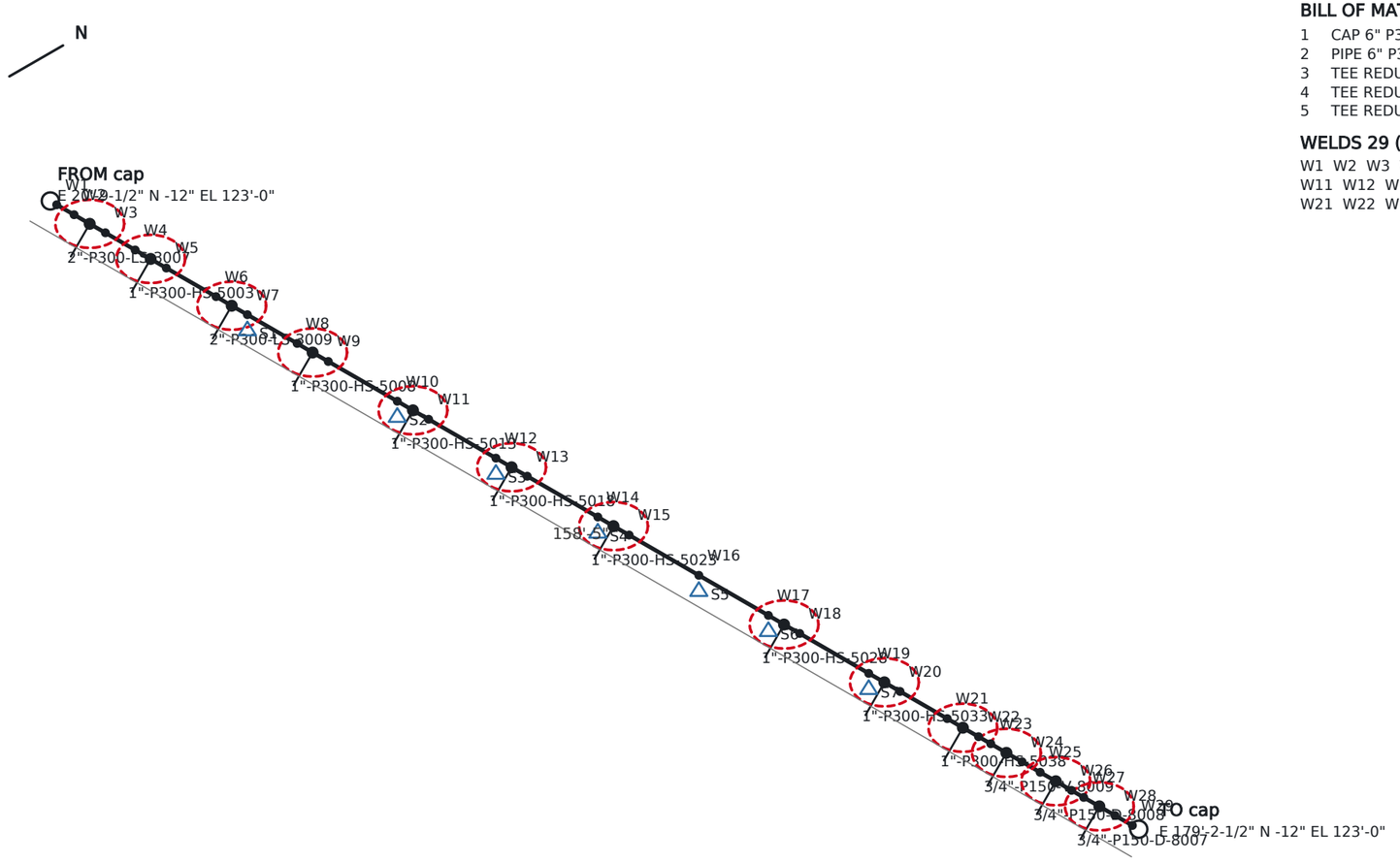
TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-500-02-TML01	6"-P300-LS-2003/T:2"-P300-LS-3007	Tee 6"-P300-LS-2003/T:2"-P300-LS-3007 branch outlet	E 22'-0" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML02	6"-P300-LS-2003/p4	Pipe 6"-P300-LS-2003/p4 mid-length	E 23'-9" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML03	6"-P300-LS-2003/T:1"-P300-HS-5003	Tee 6"-P300-LS-2003/T:1"-P300-HS-5003 branch outlet	E 25'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML04	6"-P300-LS-2003/p6	Pipe 6"-P300-LS-2003/p6 mid-length	E 30'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML05	6"-P300-LS-2003/T:2"-P300-LS-3009	Tee 6"-P300-LS-2003/T:2"-P300-LS-3009 branch outlet	E 35'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML06	6"-P300-LS-2003/p8	Pipe 6"-P300-LS-2003/p8 mid-length	E 40'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML07	6"-P300-LS-2003/T:1"-P300-HS-5008	Tee 6"-P300-LS-2003/T:1"-P300-HS-5008 branch outlet	E 45'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML08	6"-P300-LS-2003/p10	Pipe 6"-P300-LS-2003/p10 mid-length	E 55'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML09	6"-P300-LS-2003/T:1"-P300-HS-5013	Tee 6"-P300-LS-2003/T:1"-P300-HS-5013 branch outlet	E 65'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML10	6"-P300-LS-2003/p12	Pipe 6"-P300-LS-2003/p12 mid-length	E 75'-0" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML11	6"-P300-LS-2003/T:1"-P300-HS-5018	Tee 6"-P300-LS-2003/T:1"-P300-HS-5018 branch outlet	E 84'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML12	6"-P300-LS-2003/p14	Pipe 6"-P300-LS-2003/p14 mid-length	E 95'-0" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML13	6"-P300-LS-2003/T:1"-P300-HS-5023	Tee 6"-P300-LS-2003/T:1"-P300-HS-5023 branch outlet	E 105'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML14	6"-P300-LS-2003/p16	Pipe 6"-P300-LS-2003/p16 mid-length	E 115'-8-1/2" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML15	6"-P300-LS-2003/p17	Pipe 6"-P300-LS-2003/p17 mid-length	E 135'-3-1/2" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML16	6"-P300-LS-2003/T:1"-P300-HS-5028	Tee 6"-P300-LS-2003/T:1"-P300-HS-5028 branch outlet	E 145'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML17	6"-P300-LS-2003/p19	Pipe 6"-P300-LS-2003/p19 mid-length	E 155'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML18	6"-P300-LS-2003/T:1"-P300-HS-5033	Tee 6"-P300-LS-2003/T:1"-P300-HS-5033 branch outlet	E 165'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML19	6"-P300-LS-2003/p21	Pipe 6"-P300-LS-2003/p21 mid-length	E 170'-0" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML20	6"-P300-LS-2003/T:1"-P300-HS-5038	Tee 6"-P300-LS-2003/T:1"-P300-HS-5038 branch outlet	E 174'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML21	6"-P300-LS-2003/T:3/4"-P150-V-8009	Tee 6"-P300-LS-2003/T:3/4"-P150-V-8009 branch outlet	E 175'-6" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML22	6"-P300-LS-2003/T:3/4"-P150-D-8008	Tee 6"-P300-LS-2003/T:3/4"-P150-D-8008 branch outlet	E 177'-0" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____
WO-500-02-TML23	6"-P300-LS-2003/T:3/4"-P150-D-8007	Tee 6"-P300-LS-2003/T:3/4"-P150-D-8007 branch outlet	E 178'-0" N -12" EL 123'-0"	0.21875 in	0°: ____	90°: ____	180°: ____	270°: ____

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Gauge calibrated on a step block; operator qualification checked	Owner UT procedure	Calibration within the owner's tolerance	Calibration record	R	R	R				
2	Readings at 23 TMLs	SW-WO-6	Four readings per TML recorded	Thickness record per TML	S	S	S				
3	Results reviewed against the owner's minimum wall	Owner integrity procedure	Every reading above the minimum, or a work order raised	Signed thickness report	R	H	R				
4	Insulation and access reinstated; work order handed back	SW-ITP-3	All activities accepted	Signed ITP	H	H	R				

ISOMETRIC — 6"-P300-LS-2003 — Ultrasonic thickness survey of 6"-P300-LS-2003



BILL OF MATERIALS

1	CAP 6" P300	2
2	PIPE 6" P300 BEVELLED ENDS	147'-2"
3	TEE REDUCING 6" × 2" P300	2
4	TEE REDUCING 6" × 1" P300	8
5	TEE REDUCING 6" × 3/4" P300	3

WELDS 29 (0 FIELD) · SUPPORTS 7 · SPOOLS 1

W1 W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15 W16 W17 W18 W19 W20
W21 W22 W23 W24 W25 W26 W27 W28 W29

6"-P300-LS-2003

CLASS S300 — Low-pressure steam — INSULATED
SERVICE LS · 6" · P300 · DESIGN 300 psi
cap → cap
PIPE 147'-2" · WEIGHT 2144 lb

DWG ISO-6"-P300-LS-2003 REV 0 WO-500-02

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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